

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. You may recall that we recently learned the derivative of $y = \sec(x)$.
 - a) Without referencing your notes or other sources attempt to prove/derive the derivative rule for $y = \sec(x)$.

Summary Point

- b) What are some of the benefits of knowing how to derive a derivative rule versus just having it memorized? What are some of the benefits of having it memorized versus always having to derive it?

2. Show that if $f(x) = \frac{\sin(x)}{1 + \cos(x)}$ then $f'(x) = \frac{1}{1 + \cos(x)}$.

3. Determine the equation of a line which is perpendicular to $y = \frac{\tan(x)}{1 + \tan(x)}$ at $x = \frac{\pi}{4}$.

4. Suppose that $f(x) = \csc(x)$. For what values of x will the slopes of tangent lines on $y = f'(x)$ be horizontal on $[0, 2\pi]$.

5. Evaluate $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + h\right) - \sin\left(\frac{\pi}{3}\right)}{h}$.

6. Determine all values of x at which the slopes of tangent lines on $f(x) = \sin(x)$ and $g(x) = \cos(x)$ are opposite values.

Summary Point

7. Show that the function $f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ is differentiable at $x = 0$. *Hint:* You need to use the definition of the derivative.